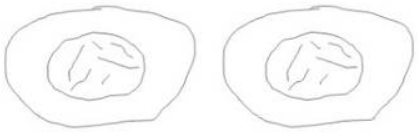
Type MS · Read two values from a graph, chart or table and find the difference

Answer 1 of 13 · Question 2 part a.i · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

2 (a) (i)	156 – 72; 84;	ECF if candidates use incorrect values from the graph with correct answer from for incorrect values = 1 mark	2
(ii)	**C** 9 minutes;		1
(iii)	In the following order only: • oxygen; • diffusion; • aerobic/cellular; • decreases/reduces/goes down;		4
(b) (i)	 must have 6 chromosomes shape not important	One mark for each correct cell drawn	2
(ii)	**B** metaphase; A is incorrect as chromosomes do not align at the equator during anaphase C is incorrect as chromosomes do not align at the equator during prophase D is incorrect as chromosomes do not align at the equator during telophase		1
			Total 10

Type MS · Read two values from a graph, chart or table and find the difference

Answer 1 of 13 · Question 2 part a.i · 2 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

uestion number	Answer	Notes	Mark s

Type MS · Read two values from a graph, chart or table and find the difference

Answer 2 of 13 · Question 5 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

5(a)	• correct labels on correct axes; • correct/appropriate scale; • correct plotting; • suitable lines; • lines labelled;	Allow line of best fit	**(5)**

Question Number	Answer	Additional guidance	Mark
5(b)	12.5 - 2.5; 10;	ECF from 5a	**(2)**

Question Number	Answer	Additional guidance	Mark
5(c)	Any three from: • exercise requires more energy; • from respiration; • more heat released raising body temperature; • more sweat released to cool body;	Reject energy produced/made Ignore anaerobic respiration	**(3 max)**

Total 10 Marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 2 of 13 · Question 5 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Read two values from a graph, chart or table and find the difference

Answer 3 of 13 · Question 4 part b.i · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

4 (a)	• IUD; • hormonal/pill; • barrier/named;		3
(b) (i)	4.35 - 0.65; 3.7 mm;		2
(ii)	• increase in thickness/width; • if fertilisation occurs; • allows implantation to occur;		3
(iii)	days 0 - 6;		1
(iv)	• endometrium line would remain high/level off/plateau; • oestrogen line would level off; • at a lower level;		3

Total 12 marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 3 of 13 · Question 4 part b.i · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 4 of 13 · Question 6 part a.i · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
6	(a)	(i)	18-12; 6;	accept 17.8/17.9	2
		(ii)	Two from: • volume of milk; • volume of the starter culture/live yoghurt; • type of starter culture • type of milk; • type of live yoghurt; • mixing;	reject 'amount'	2
		(iii)	• repeat the investigation at more temperatures; • between 20°C and 60°C;		2
		(iv)	• heated to kill (unwanted) bacteria in the milk; • cooled so that the bacteria in the starter culture are not killed;		2
	(b)		Any two from: • pH 4.4 inhibits the bacteria/bacterial growth; • denatures enzymes; • changes the shape of the active sites • lactose cannot bind to the enzyme/no enzyme-substrate complexes formed;		2
	(c)	(i)	Any two from: • the beads can be re-used; • product doesn't need to be treated to remove the enzyme; • large surface area to increase the rate of reaction; • enzyme more temperature stable;		2
		(ii)	• long/villi/microvilli to increase surface area; • concentration gradient maintained with blood flow; • short diffusion distance;		3

Total 15 marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 4 of 13 · Question 6 part a.i · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 5 of 13 · Question 6 part b.i · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
6	(a)	(i)	Three from: • the use of embryonic stem cells is not needed/embryos not destroyed; • easily available source of cells/easy to obtain/no need to find a donor; • same cell/tissue/blood type of the patient/same antigens; • reduced chance of rejection/not recognised as foreign;		3
		(ii)	Two from: • unlicensed clinics could be used; • stem cells could develop into tumours/cancer; • risk of rejection (if the cells are not the recipients); • increased risk of infection;		2
	(b)	(i)	860 000 and 480 000; 380 000;	Ecf allow one mark for incorrectly derived figures from graph but correct answer from these figures	2
		(ii)	**D** structure Z;		1
	(c)		Three from: • cut out desired/beta-carotene/gene/gene for vitamin A; • using/cut with restriction enzymes; • insert into plasmid; • insert into/infect rice genome; • to produce golden rice;		3

Total 11 marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 5 of 13 · Question 6 part b.i · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 6 of 13 · Question 3 part a.i · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

3 (a) (i)	• 120 - 5; • = 115 mm Hg;	allow +/- 2 for each figure ECF	2
(ii)	A; (arteries) B/C/D are represented by other parts of the graph		1
(iii)	D; (veins) A/B/C are represented by other parts of the graph		1
(b)	any 5 from • heart; • (left) ventricle contracts; • forces blood out of heart/through aorta; • high pressure; • ventricle relaxes; • pressure falls;		5
(c)	• capillaries; • no pulse/fluctuations; • as no muscle/elastic tissue; • (blood only pushed along at) low pressure/pressure drops;		4

Total 13 marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 6 of 13 · Question 3 part a.i · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 7 of 13 · Question 4 part c.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

Number				
4	(a)	any two from • hearing/sound; • touch; • taste; • smell;		2
	(b) (i)	any two from • as thickness decreases, focal length increases; • interval between focal lengths increases as thickness decreases; • (constant when two variables multiplied hence) inversely proportional;		2
	(ii)	diameter decreases;		1
	(iii)	any four from • rays from near object diverging; • ciliary muscle contracts; • suspensory ligaments slacken/relax; • so lens becomes thicker/more convex; • more powerful lens/light refracted more/increased refraction; • (allows it to focus because) focal length becomes shorter/decrease;		4
	(c) (i)	between 575 and 580 nm;		1
	(ii)	• 400 - 530; • 130 nm;	ECF	2
	(iii)	• greater range of wavelengths can be seen/absorb light of different wavelengths; • gives better colour perception/vision/differentiation/can see different colours;		2

Total 14

Type MS · Read two values from a graph, chart or table and find the difference

Answer 7 of 13 · Question 4 part c.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 8 of 13 · Question 4 part b.ii · 2 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
4	(a)	(i)	• glucose; • carbon dioxide and water;	Reject formulae	2
		(ii)	aerobic;		1
		(iii)	B; (mitochondrion)		1
		(iv)	ATP/adenosine triphosphate		1
	(b)	(i)	Any two from • more people/repeat tests: • different gender/diet/location/age; • carry out test over more years/longer time period; • calculate the mean/ average;		2
		(ii)	• 7250 - 6550;	ecf if wrong figures used in a subtraction calculation but correct answer 1 mark	
			• = 700kJ;		2
		(iii)	Any five from • metabolic rate increases in colder temperatures; • correct comparative reading from graph e.g. metabolic rate is higher in January compared to July; • (in colder temperatures) increased rate of respiration; • releases heat (energy)/more energy needed; • (energy needed) to maintain body temperature/reference to keeping body warm; • (in warmer weather) more heat lost (by radiation / convection/sweating);	ORA for mps 1, 3, 4, 5, 6 Allow marking point if correct figures given from graph	5
		(iv)	uptake of oxygen/carbon dioxide output;	Allow amount of glucose used	1

Total 15

Type MS · Read two values from a graph, chart or table and find the difference

Answer 8 of 13 · Question 4 part b.ii · 2 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 9 of 13 · Question 3 part b.i · 1 mark

2023 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

3 (a)	pathogens;		1
(b) (i)	1.2 years;		1
(ii)	• active; • artificial;		2
(iii)	Any two from • weakened/dead/attenuated organisms; • toxins; • antigens; • RNA;	Allow named bacterium/virus Ignore genetic material	2
(c)	Any two from • passed from mother; • during breastfeeding/from breast milk; • across placenta;	Ignore umbilical cord	2
(d)	Any five from • antibody level falls; • below safe level/to unsafe level; • greater risk of disease; • booster dose increases antibody levels/remains above unsafe level; • person immune/can't catch disease; • increased number of memory cells;	Allow faster response	5
(e)	Any two from • antibody level higher; • for longer period/decreases more slowly/remain for many years; • levels of antibody increase faster after booster;		2

Total 15 marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 9 of 13 · Question 3 part b.i · 1 mark

2023 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 10 of 13 · Question 7 part b.i · 2 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

7 (a)	(i)	• volume/concentration of enzyme/lipase; • volume/concentration of indicator; • volume/same type of olive oil; • volume/mass of bile saltes	Ignore amount for all mps	3
	(ii)	40°C;		1
	(iii)	no values given between 30°C and 40°C; no values given between 40°C and 50°C; optimum could be between temperature intervals given/below/above 40ºC/50ºC;		3
(b)	(i)	• tube 3 = 11.5 hours and tube 8 = 24 hours; • 24 - 11.5 = 12.5 hours;	Ecf incorrect values in a subtraction calculation with a correct answer 1 mark	2
	(ii)	• lipase digests/breaks down olive oil; • produces fatty acids; • these acids lower the pH;		3
(c)		• no bile salts so no emulsification of oil/oil not broken down to droplets; • less surface area (for lipase to work on); • lipase only working very slowly/not very active/takes longer to digest oil;		3

Total 15 marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 10 of 13 · Question 7 part b.i · 2 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · Read two values from a graph, chart or table and find the difference

Answer 11 of 13 · Question 1 part c.ii · 2 marks

2024 JUN · Paper 02 · find part c.ii in this mark scheme extract

Number			
1 (a)	<table><tr><th>Disease</th><th>Type of pathogen</th></tr><tr><td>cholera</td><td>Bacterium/bacteria;</td></tr><tr><td>AIDS</td><td>virus;</td></tr><tr><td>athlete's foot</td><td>Fungus/fungi;</td></tr></table>	Reject more than one line from any disease	(3)
(b)	In the following order: bacterium; sexual intercourse; barrier; antibiotics;		(4)
(c) (i)	• overall decrease; • fluctuates • steep decreases from 1990;		(2)
(ii)	17 - 9; 8;	Allow tolerance of +-0.5 ecf for mp2	(2)
(d)	C antigens,white blood cells produced,antibodies destroy pathogens; The following are incorrect because - A vaccinations do not contain antibodies B platelets are not produced following a vaccination D vaccinations do not produce platelets		(1)

Total for Question 1 = 12 marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 12 of 13 · Question 2 part a.ii · 2 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number				
2	(a) (i)	• axes labelled with units; • axes correct way round and suitable scale; • points plotted correctly;; • straight lines joined; • lines identified;	time without with 0 65 110 30 120 215 60 130 230 90 95 200 120 80 190 150 65 180 180 65 170 don't include in published version	6
	(ii)	• without = 65 mg per 100cm^3; • with = 120 mg per 100cm^3;		2
	(iii)	any three from • without, level starts lower/level lower throughout; • smaller difference between highest and lowest values; • rises to lower levels/rise less quickly; • returns to normal (quicker)/same level 150-180;	ORA for person with diabetes	3
	(b)	• sample in a test/boiling tube; • add Benedict's reagent/solution; • heat in a water bath; • wear goggles; • brick red colour if glucose present; if candidate uses glucose testing sticks • sample in a test/boiling tube; • use of testing stick; • dip into urine; • compare colour with chart; • use of goggles;	Accept yellow/green/orange	5

Total 16 marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 12 of 13 · Question 2 part a.ii · 2 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Read two values from a graph, chart or table and find the difference

Answer 13 of 13 · Question 7 part a.iii · 2 marks

2022 JAN · Paper 01 · find part a.iii in this mark scheme extract

Question number					
7	(a)	(i)	carbon hydrogen and oxygen;		1
		(ii)	nitrogen	Allow sulfur	1
		(iii)	19.6 - 12; 7.6 g	Allow ECF	2
		(iv)	protein 1.2 g × 17 = 20.4 kJ; lipid 4.8 g × 37 = 177.6 kJ; total = 198 kJ;	Allow ECF for 2nd and 3rd mark	3
	(b)		add alcohol to fish; add water; cloudy/milky appearance/emulsion indicates lipid;		3

Total 10 marks

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